

Where Does Jagged Competence Come From?

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Abstract

Capable systems often show jagged competence: low average error alongside failures on particular inputs. We ask where it comes from in a task built from two known layers. A lower layer A computes five per-slot sums from records; an upper layer B uses the slot-1 sum and a mode carried over from earlier boards to predict the next board’s category, so B cannot be computed from the current A alone: B is a strict extension of A. We train small recurrent networks on B and measure, against exact ground truth, what they acquire of A. The central finding is that learning B gives the network a jagged version of A, measured through probe readability and supervised outputs. It is readable where B needs it (slot 1, 97–98% by a linear probe) and becomes less readable where B does not (the other slots, 5–10%); category misreads concentrate near the slot-1 cutoffs; and when A is trained explicitly it comes out only approximately right. The networks show jagged competence in B: natural KL below 3×10^{-4} bits coexists with a maximum law TV of about 0.25 on constructed histories. Matched experiments show that even a good A is not enough: connecting a learned A to B cuts misreads 1.3–17-fold, the same exact A gives fewer misreads as one-hot inputs (0–2) than as numerical inputs (10–317), one B update makes an exact A inexact unless the B gradient is blocked, and exact A still leaves some B failures. In this task, uneven acquisition, access and preservation of the lower theory explain part of the jagged competence of a network that learns the theory built on it; some failures remain unexplained. Whether the same holds in larger systems is a test to run.

Keywords. jagged competence; controlled task; recurrent network; recoverability; prediction; preservation; calibrated interventions.

1 Introduction

Jagged competence combines strong performance with failures on particular inputs.¹ Field experiments document an uneven AI frontier [5], and Gans [8] models how average quality can hide local unreliability. The pattern is easy to observe and hard to explain: in a large system we rarely know which intermediate quantities the task requires, or whether the system computes, uses and keeps them, and without that knowledge “jagged” remains a description.

We study the question in a task built from two known layers. Layer A computes five per-slot sums from a board of records. Layer B uses the slot-1 sum’s category and a mode carried over from earlier boards to set the probabilities of the next board’s category. The same board can need different predictions after different histories, so B cannot be computed from the current A alone; in the terms of Six Birds Theory, B is a strict extension of A (section 3.2). A working hypothesis of that programme, motivated by its picture of theory levels [26, §1.1] rather than proved for trained

¹Gans [8] credits the term “jagged intelligence” to Andrej Karpathy; we use “competence” for performance in this task.

networks, is that a network trained on an upper theory comes to know the theory beneath it. Because both layers are known exactly, we can test this: train networks on B only, and measure what they acquire of A and how it relates to where B fails.

The answer, in this task, is that learning B gives the network a jagged version of A, and that this explains part of the jagged competence in B. The network makes readable the part of A that B needs while the rest becomes less readable; in r12’s uniform-sampling census, category misreads concentrate near the slot-1 cutoffs; and when A is trained explicitly, it comes out largely but not exactly right. The networks reach a natural KL below 3×10^{-4} bits, more than thirty times below our 0.01-bit bar, yet fail on constructed histories (maximum law TV 0.25–0.26), and their category misreads sit near the slot-1 cutoffs (sections 5.1 and 5.2). Here “acquire” means what we measure, readable by probes and approximately computed when supervised; we do not demonstrate an internal layer, and the jagged A explains part of B’s failures, not all of them.

A second lesson is that knowing an intermediate quantity is not one thing. A network can hold A in readable form without established exact computation, compute it without delivering it to the prediction, deliver it in a form that supports less reliable prediction, or lose it under further training; and even an exact, delivered A leaves some failures in B. These are separate achievements, not a required sequence, and matched comparisons isolate the effects of connection, encoding and prediction gradients (sections 5.3 to 5.7).

Contributions.

- The finding that learning a strict extension gives a network only a jagged version of the lower theory, and that this explains part of the upper one’s failure, with the unexplained cases stated (sections 5 and 6).
- A decomposition of what it means for a network to know an intermediate quantity into five measurable achievements (recoverability, computation, access, preservation and reliable prediction), with matched experiments on access, encoding, preservation and target noise (section 5).
- A layered task with exhaustive board ground truth and calibrated measurements (sections 3 and 4), and a provided ledger and code rebuilding every reported number, figure and table (section F).

2 Related work

Jagged competence. Dell’Acqua et al. [5] document an uneven AI frontier in a field experiment, and Gans [8] models uneven local reliability through gaps in knowledge coverage. Kalai and Vempala [15] derive calibration-related bounds on errors about arbitrary facts, while distinguishing systematic facts. Our task has a systematic rule, so those bounds do not explain its errors.

What training makes available. The training task shapes which input features a network represents and how readable they are [12], and gradient descent can favor simple features and neglect others [23, 21, 11]. We ask which parts of a designed intermediate computation become recoverable under final-output supervision, and whether recoverability accompanies computation, access and preservation.

Probing versus computation and use. Probe accuracy alone does not show that a network uses what the probe finds: probes need controls for their own capacity [13, 2], high probing accuracy can coexist with task irrelevance [22], and removing information tests use more directly [6]. Darade

and Thorat [3] recover candidate arithmetic intermediates but do not find them transmitted along the tested causal routes, and Basu et al. [1] report a dissociation between probe discrimination and output behavior. Interchange interventions test whether an aligned internal variable plays a causal role [9, 10], and causal tracing and model editing locate and modify stored associations [19]; our swap test follows this design but failed selectivity under the tested alignment, a limit of the test rather than evidence of absence. We separately measure exact output computation and predictive access, then test whether connecting a learned sums output reduces errors.

Controlled tasks with known ground truth. Controlled tasks allow exact comparison of internal states and behavior with a reference: board-game world states [17], the gap between next-token accuracy and a coherent world model [29], extrapolation on formal languages [4], synthetic reasoning [30], grokking and superposition [20, 7]. Shintani [24] separates arithmetic failures into stages tested with matched data interventions. We share this controlled-decomposition approach, while studying a recurrent layered task with measures of recoverability, output computation, access and preservation.

Auxiliary targets, soft targets and interference. Soft targets carry more information than sampled labels [14], objectives can conflict [31], and training on a new objective can degrade an earlier one [18, 16]. Shu and Wei [25] propose an account of jaggedness involving uneven optimization allocation and capability coupling. Our erosion experiment establishes dependence on the prediction gradient, but does not identify interference as its mechanism.

3 Task and setup

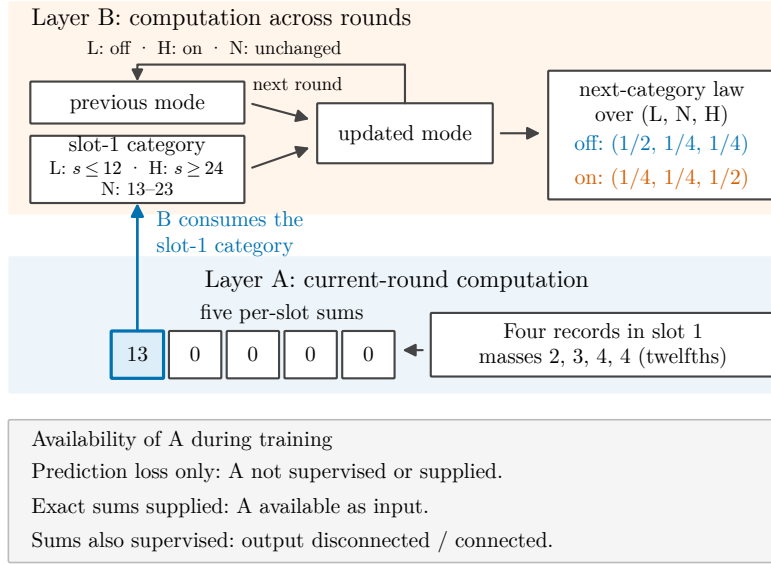
In the records game, the quantities a network might compute, the hidden variable and the correct next-step probabilities are known exactly for every input, and all boards can be enumerated (fig. 1).

3.1 The game

Boards and sums. A *record* places one mass in one of five *slots*. The six masses are $1/6$, $1/4$, $1/3$, $1/2$, $2/3$ and $3/4$, written below in integer twelfths (2, 3, 4, 6, 8, 9). A *board* is an ordered sequence of four records. Its *sums* are the five per-slot totals, each one of the 36 attainable values 0 and 2–36 twelfths. Computing the sums defines layer A. The $30^4 = 810,000$ ordered placements yield 24,435 distinct *completed sum boards*; many record sequences realize the same completed sum board, and we call one such realization a *rendering*.

Category, mode and law. Only slot 1 matters for prediction. Its sum gives the board’s *slot-1 category*: L when the sum is at most 12 twelfths, H when it is at least 24, and N otherwise (13–23). The task keeps a binary *mode*, which starts off. An H board switches it on, an L board switches it off, and an N board leaves it unchanged. The next board’s category is drawn from the *law* $(\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$ over (L, N, H) when the mode is off and $(\frac{1}{4}, \frac{1}{4}, \frac{1}{2})$ when it is on. Given the category, the next board is a uniformly chosen completed sum board of that category, rendered as a legal placement of records. At every round the network sees the current board’s records and outputs a *prediction*: a probability vector over the next board’s category. Predicting the law is the upper task layer B. In the *equal-law control* both modes use $(\frac{3}{8}, \frac{1}{4}, \frac{3}{8})$, so the mode is irrelevant; it checks that the measurements do not report mode use where none is possible.

(a) Exact task construction



(b) Same sums, different laws

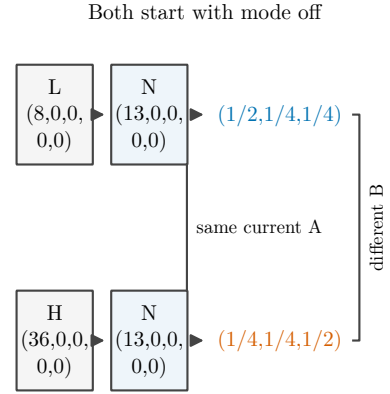


Figure 1: Two task layers and their history witness. (a) A maps four records to five per-slot sums. B uses their slot-1 category and the preceding mode to update the mode and determine the next-category law. The annotations identify training conditions, not evidence that a network learned the depicted decomposition. (b) The histories end on the same neutral sum board but retain different modes and require different laws. Values are integer twelfths; law entries are ordered L/N/H. The task has 24,435 completed sum boards and 1,555 local histories. Everything shown is exact task ground truth, not model output.

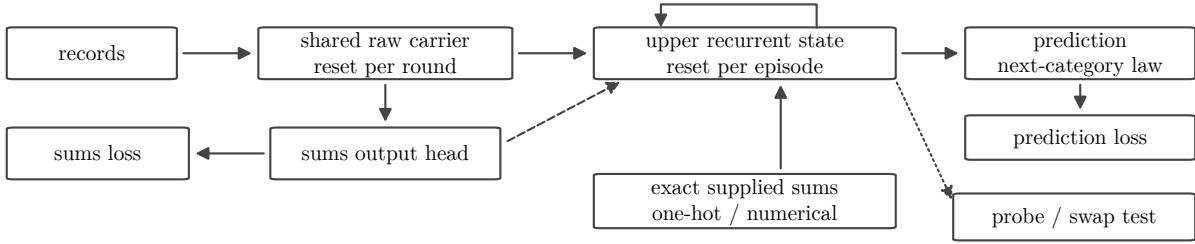
Boards near the cutoffs. 22,491 completed sum boards are L, 1,835 are N and 109 are H. The *cutoff band* (slot-1 sums 11–13 and 23–25; 930 boards) is 2.5% of the L boards, 16.9% of the N boards and 45.9% of the H boards. Under uniform sampling within a category, these shares are also the training probability of a band board; the rarity conditions of r11 and r12 change them.

3.2 Two task layers

Layer A maps each round’s records to its five sums. Layer B operates over sequences of these completed sum boards: the slot-1 category updates the preceding mode, and the updated mode determines the next-category law (fig. 1a). The exact update needs only two things, the current board’s slot-1 category and the previous mode. A network that predicts reliably must therefore combine information about the current board’s category with information from earlier boards; it need not do so through the exact sums.

Six Birds Theory gives this relationship a precise meaning. A description B *strictly extends* a description A when B draws distinctions that no function of A can draw. The theory uses this as its certificate of novelty: a new level of description adds predicates that are not definable from the old one, rather than computing further within it [26, §1.1]. Formally, B does not factor through A, which holds exactly when some pair of cases has equal A and different B (27, Theorem 12; 28, §3.5); this non-factorization is the only condition we use. Under the original law, B (the next-category law) strictly extends the current A description (the five sums): no function of the current sums gives the correct law for every history. Two histories ending on the same neutral board have identical sums but can retain different modes and require different laws (fig. 1b). B is determined by the sequence of A objects and the initial mode, not by the current A object alone. Neither containment

(a) Recurrent architecture and the optional sums connection



(b) Route and loss matrix

version / condition	records route	sums route	loss
records $\rightarrow$ prediction	yes	masked	prediction loss
records+sums $\rightarrow$ prediction	yes	exact frozen	prediction loss
records $\rightarrow$ prediction+sums	yes	disconnected	prediction + sums loss
Connected (frozen)	yes	fixed sums output	prediction + sums loss
Connected (live)	yes	live sums output	prediction + sums loss
Numerical encoding	yes	exact numerical	prediction loss
Probability targets	base version	as the base version	base losses; exact law

Figure 2: Recoverability, computation and access are distinct measurements, not interchangeable routes. (a) The sums output head and its loss attach to the shared carrier; a dashed optional connection feeds sums output to prediction in the connected variants. (b) The matrix identifies the routes and losses in the three versions and r13 variants; probability targets retain their base version’s routes and replace sampled targets with the exact law. The probe fits a reader on frozen states; the swap replaces an aligned donor block, with selectivity checked separately. No measurement, seed average or control result is implied by the icons; procedures and their calibration are in Appendix C.

nor recovery of A from B is required. The equal-law control removes this distinction.

These are the task’s layers. Whether training realizes their decomposition internally is the empirical question.

3.3 Network, versions and training

Network. For the original task all networks share one architecture (fig. 2); in r10 the upper state also receives the public query. Within a round, a one-layer GRU of width 16 reads the four records and produces the *raw carrier*, which is reset at every round. Across rounds, an upper GRU cell of width 32 receives the raw carrier and, where a version provides them, the five sums as one-hot vectors; a linear head on the upper state gives the prediction. Supplied sums come from a separately trained sums network that is exact on all 1,555 local histories (every ordered sequence of up to four masses in one slot) and is kept frozen. Details are in section F.1.

Versions. Each version name states the input, an arrow, and what the loss is computed on. In **records $\rightarrow$ prediction** the input is the records and the loss is on the prediction only. In **records+sums $\rightarrow$ prediction** the input also includes the exact one-hot sums. In **records $\rightarrow$ prediction+sums** the input is the records and the loss is on the prediction plus a *sums output*, a linear head on the raw carrier trained to give the five sums and not fed to the prediction. r8

trained the first two versions and a third, **records+sums** $\rightarrow$ **prediction (sums route only)**, whose upper state receives only the exact sums; records $\rightarrow$ prediction+sums first appears in r10. The versions vary the supervision of A and its availability to B: no explicit sums supervision, exact supplied sums, or supervised sums outputs disconnected from or connected to prediction. Further variants change one factor. **Connected (frozen)** and **Connected (live)** feed the sums output of a trained records $\rightarrow$ prediction+sums network into the prediction, from a fixed copy or from the sums output being trained. **Numerical encoding** gives the exact sums as $(s/36, 1 - s/36)$ per slot instead of one-hot. **Probability targets** train against the exact law instead of one sampled next category. The **erosion continuation** starts from an exact sums network that is no longer frozen and updates either the prediction loss only or the prediction plus sums loss.

Training. Full training runs use three seeds (0, 1, 2), a fixed 20,000 updates, AdamW (learning rate 0.003, weight decay 0.01, batch 256), float32 and freshly drawn episodes; no checkpoint is selected by accuracy. From r12 on, each batch also contains *history coverage*: constructed prefixes of an L or H board followed by up to eight N boards. The r13 erosion replays instead run for 200 updates and compare three optimizers for the sums network (AdamW, AdamW with a tenfold smaller learning rate, and SGD matched to the norm of AdamW’s first gradient-only displacement) with a control that blocks the prediction gradient to the sums network while weight decay continues. r9 trains nothing. Details are in section F.1.

3.4 Study history and comparisons

The evidence comes from six studies, r8–r13, each registered with its endpoints and criteria before it ran. r8 trained the three r8 networks under both laws. r9 diagnosed the saved r8 networks. r10 retains A but changes B to the *all-slots task*, testing readability and use of all five sums: there each slot remembers its last nonzero sum (initially zero) and the law over the next board’s total sum modulo three depends on the remembered value of a publicly queried slot, not its current sum (36 distinct laws). r11 varied board rarity near the cutoffs, the amount of sums supervision, and the erosion continuation. r12 added history coverage. r13 ran four mechanism experiments: target noise, access, encoding and erosion. Follow-ups, corrected audits and supplementary analyses are labeled in section A; the campaign was not one confirmatory study.

A comparison is *matched* when the code, data streams, initialization, number of updates and evaluation panels are identical and only the stated factor differs. All other comparisons are *historical*. Comparisons between r11 and r12 change the curriculum and the loss weighting at once; comparisons between r10 and the other studies change the task. The r13 access and encoding experiments share one reference condition, the exact one-hot supplied sums, which we count once.

4 Measurements

Each reported effect is read against a floor and a ceiling. For prediction, the ceiling is the exact law (zero KL and zero law error) and the main floor is the *board-only predictor*, which knows the current board exactly but not the history. For probes, the floors are the majority class, shuffled labels and the network at update 0, and the ceiling is the same probe fitted on an exact sums carrier.

Prediction: average and constructed tests. *Natural KL* is $\text{KL}(\text{law} \parallel \text{prediction})$ in bits, averaged over 60,000 positions (5,000 test episodes of 12 rounds); *unseen KL* is the same quantity on test positions whose preceding pair of boards never occurred in training. Average error can be

low while specific histories fail, so we also score *law panels*: constructed histories made of an L or H anchor board followed by k N boards, $k = 0, \dots, 8$, with 32 variants per anchor and length (576 histories). The *law TV* is the total-variation distance between the prediction and the law at the end of a history. The *short panel* covers the anchors and N^1 – N^2 ; the *long panel* covers N^3 – N^8 .

Passing. A run *passes* when it meets every prediction criterion registered for its study at the fixed 20,000-update endpoint. For the original task under the original law these are: natural and unseen KL at most 0.01 bits; maximum law TV at most 0.02; witness recovery (the share of the history effect captured one step after the anchor) at least 0.8; and prediction TV at most 0.02 under rerendered boards and on the registered state-intervention panels. The equal-law control and r10 use their own criteria; r9 and the erosion replays have no pass endpoint. Complete criteria for every study are in the supplement (section C.1). A pass is a statement about these criteria, not exhaustive correctness.

Census. To locate failures on boards rather than histories, we score every completed sum board after an L anchor and after an H anchor. The two predictions form the board’s *signature*. A *categorical misread* is a board whose signature is closer to the exact signature of a wrong category than to that of its own. A *TV-error count* is a board whose larger prediction TV over the two contexts exceeds 0.02. These are different measures and we never merge them: a board can miss the 0.02 bar without being misread. Misread signatures are undefined under the equal law.

Recoverability. A *probe* is a classifier fitted after training on frozen internal states, the raw carrier or the upper state, that predicts one slot’s sum. It is fitted on 2,048 boards and scored on 512 other boards per slot. We use linear probes and one-hidden-layer probes of width 64. A sum is *recoverable* when the probe scores above the majority-class and shuffled-label floors. Separately, the probe’s *gain* is its improvement over the same probe on the network at update 0, with a paired 95% interval over the evaluation boards. That interval describes probe uncertainty for one network, not variation across seeds, which we report as per-seed values.

Computation and preservation. *Local exactness* is the fraction of the 1,555 local histories for which a sums output is exactly correct. *Board-vector exactness* is the fraction of the 24,435 completed sum boards whose five-sum vector is correct; the early readouts of the r11 erosion continuation use a fixed sample of 128 boards instead. Preservation is the change in these quantities during further training from an exact start. To compare optimizers whose steps differ in size, we track *cumulative movement*: the sum over updates of the Euclidean norm of each parameter step of the sums network.

Access and use. Recoverable is not used. We say the prediction *uses* a quantity only when one of two kinds of evidence supports it: a matched comparison that changes only how the quantity reaches the prediction (connected versus disconnected, one-hot versus numerical), or a calibrated, selective swap test. In the swap test (r10), the raw carrier of the networks that read only records is mapped by a fitted orthogonal alignment onto coordinates that track the five sums, and the three-coordinate block for the queried slot is replaced with the block from a donor board that differs only in that slot’s value. *Donor-follow* is the fraction of accurate pairs for which the patched prediction is closer to the donor’s law than to the base board’s; *selectivity* requires that the other four slots’ decoded sums do not change. Calibration on a trained positive and two negatives, selectivity, and the evaluated result

Table 1: Five achievements, original law. ✓ supported; × not supported; – not established: available evidence does not establish this achievement. Fractions are seeds passing out of three, except recoverability: supported slots out of five in seed order 0/1/2. Access: *D* by design (exact sums supplied); *I* demonstrated by a matched connection intervention. Complete scope, counts and controls: Supplement Table S1.

condition	recoverability	computation	access	preservation	reliable prediction
r8: records → prediction	✓ 1/5-1/5-1/5	–	–	–	× 0/3
r10: records → prediction	✓ 5/5-5/5-5/5	–	–	–	× 0/3
r10: records → prediction+sums	✓ 5/5-5/5-5/5	–	–	–	× 0/3
r11: records → prediction (Erosion continuation)	✓ 1/5-1/5-1/5	× 0/3	–	×	× 0/3
r11: records → prediction+sums (Erosion continuation)	✓ 3/5-1/5-1/5	✓ 1/3	–	×	✓ 1/3
r12: records → prediction+sums	✓ 5/5-5/5-5/5	× 0/3	–	–	× 0/3
r12: records+sums → prediction	✓ 2/5-2/5-3/5	✓ 3/3	✓ ^{<i>D</i>}	✓ ^{<i>f</i>}	✓ 2/3
r13: records → prediction+sums	–	× 0/3	–	–	× 0/3
r13: Connected (frozen)	–	× 0/3	✓ ^{<i>I</i>}	✓ ^{<i>f</i>}	× 0/3
r13: Connected (live)	–	× 0/3	✓ ^{<i>I</i>}	–	× 0/3
r13: records+sums → prediction (exact supplied)	–	✓ 3/3	✓ ^{<i>D</i>}	✓ ^{<i>f</i>}	✓ 1/3
r13: records+sums → prediction (Numerical encoding)	–	✓ 3/3	✓ ^{<i>D</i>}	✓ ^{<i>f</i>}	× 0/3
r13: blocked prediction gradient	–	✓ 3/3 ^{<i>ℓ</i>}	–	✓ ^{<i>b</i>}	–

Note. Raw-carrier linear readers: 512 held-out boards/slot. Computation: the audited sums output, exact vectors on rendering 0 (24,435 boards); *ℓ*: 1,555 local histories. The connection intervention does not establish selective use. *f*: frozen supplier; *b*: blocked-gradient control. Frozen preservation is design-limited.

are reported separately (sections C and C.1). A swap that fails selectivity under one alignment does not show that the quantity is absent or unused; it shows only that this test cannot support use.

Calibration. Every bar was checked by running its deciding function on known positive and null cases before results were interpreted. A calibration pass shows that an instrument can discriminate; it is not a result about any network.

5 Results

Table 1 summarizes, for each version and condition, which of the five achievements of realizing the designed decomposition the evidence supports; its columns follow the order used throughout (recoverability, computation, access, preservation, reliable prediction). Unless stated otherwise, results use the original law, the 20,000-update endpoint and census rendering 0, and values are per seed in seed order (0 / 1 / 2).

5.1 Training shapes what becomes recoverable

Claim: training on B gives the network a jagged version of A, readable where B needs it and less readable where it does not. In the records → prediction networks of r8, a linear probe on the raw carrier recovers the slot-1 sum on 97–98% of held-out boards at the endpoint, up from 26–37% at update 0 (ranges over three seeds; fig. 3a). The sums of slots 2–5, which the prediction never needs, become *less* readable: from 24–37% at update 0 (mean 30%) to 5–10% at the endpoint (ranges over four slots and three seeds). A one-hidden-layer probe shows the same direction (about 60% to 38%, mean over slots 2–5 and three seeds; fig. 3b). The linear probe weights classes equally, so part of the drop below the majority floor reflects that weighting; we read the decline as reduced measured readability, not as removal of information. In the all-slots task (r10),

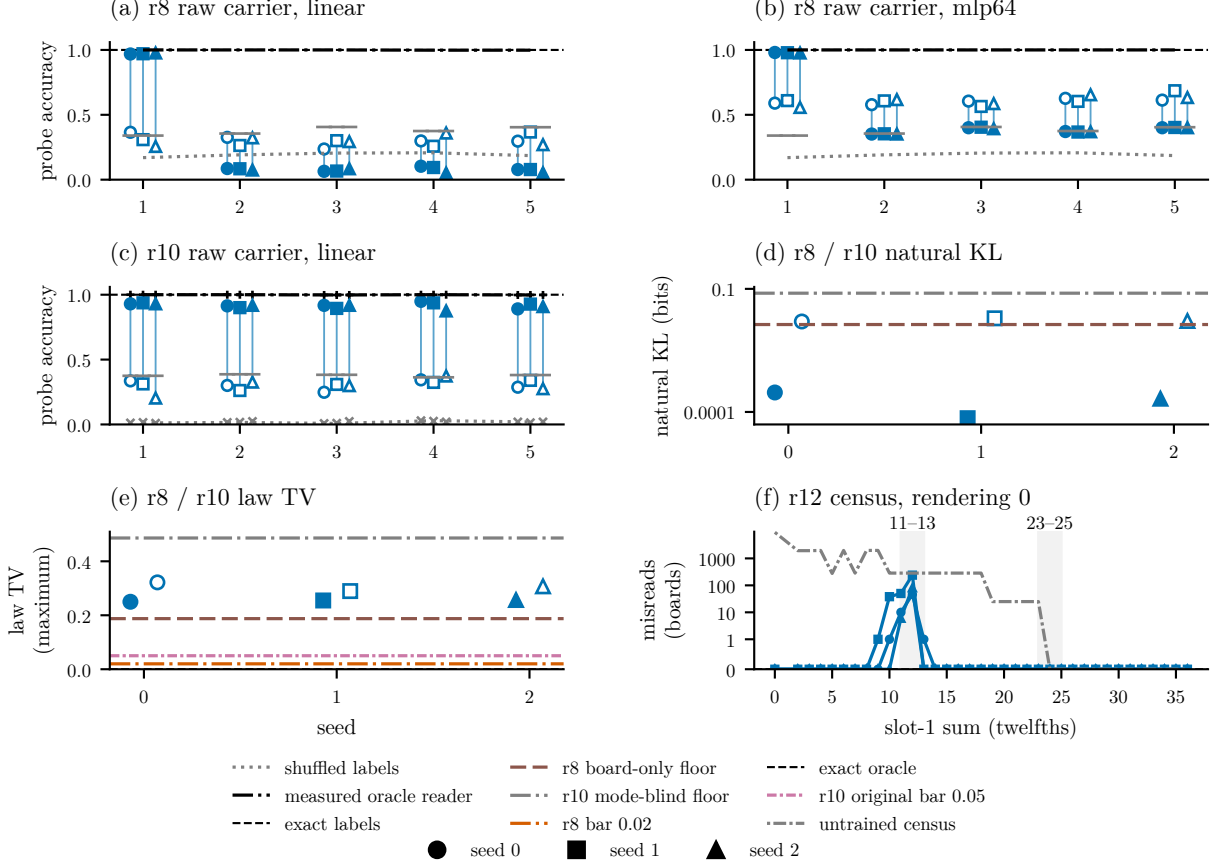


Figure 3: Recoverable sums and low average error coexist with specific failures. (a,b) r8 linear/MLP; (c) r10 linear: slots 1–5, 512 held-out boards/slot, majority ticks, shuffled floors and oracle ceiling. Open/filled: update 0/20k (a–c), r10/r8 (d,e); shapes: seeds 0/1/2. (d) Natural KL: 60,000 positions, board-only/mode-blind floors; the log axis omits the oracle’s 0 bits. (e) Law TV: 576 r8 histories/180 r10 rows, zero oracle, r8 bar 0.02 and r10 original-law 0.05 bar (its equal-law bar is 0.02). (f) r12 uniform categorical misreads: 24,435 boards, rendering 0, untrained floor, shaded cutoff bands. Compare seeds within panels; r8/r10 comparisons are historical and tasks differ. Paired probe intervals: Appendix E.

where the law depends on whichever slot is queried, linear probes on the raw carrier recover all five sums: 20–37% at update 0 and 88–95% at the endpoint over five slots and three seeds, with every paired 95% gain interval above zero (fig. 3c).

Scope. The r8–r10 comparison is historical and crosses tasks. It supports task-dependent organization of what is recoverable, not exact computation or use; rare r10 sum values (28–36) lie outside the registered scope.

5.2 Average error hides specific failures

Claim: B combines low average error with category misreads concentrated near the slot-1 cutoffs and failures on particular histories. The r8 records $\rightarrow$ prediction networks reach a natural KL of 0.00030 / 0.00007 / 0.00021 bits, far below the 0.01 bar and the board-only predictor’s 0.0135 bits (fig. 3d). Yet none of the three runs passes: on the constructed law panels the maximum law TV is 0.250 / 0.255 / 0.256, worse than the board-only predictor (0.188) and twelve times the 0.02 bar (fig. 3e). A separate census of the r12 records $\rightarrow$ prediction networks

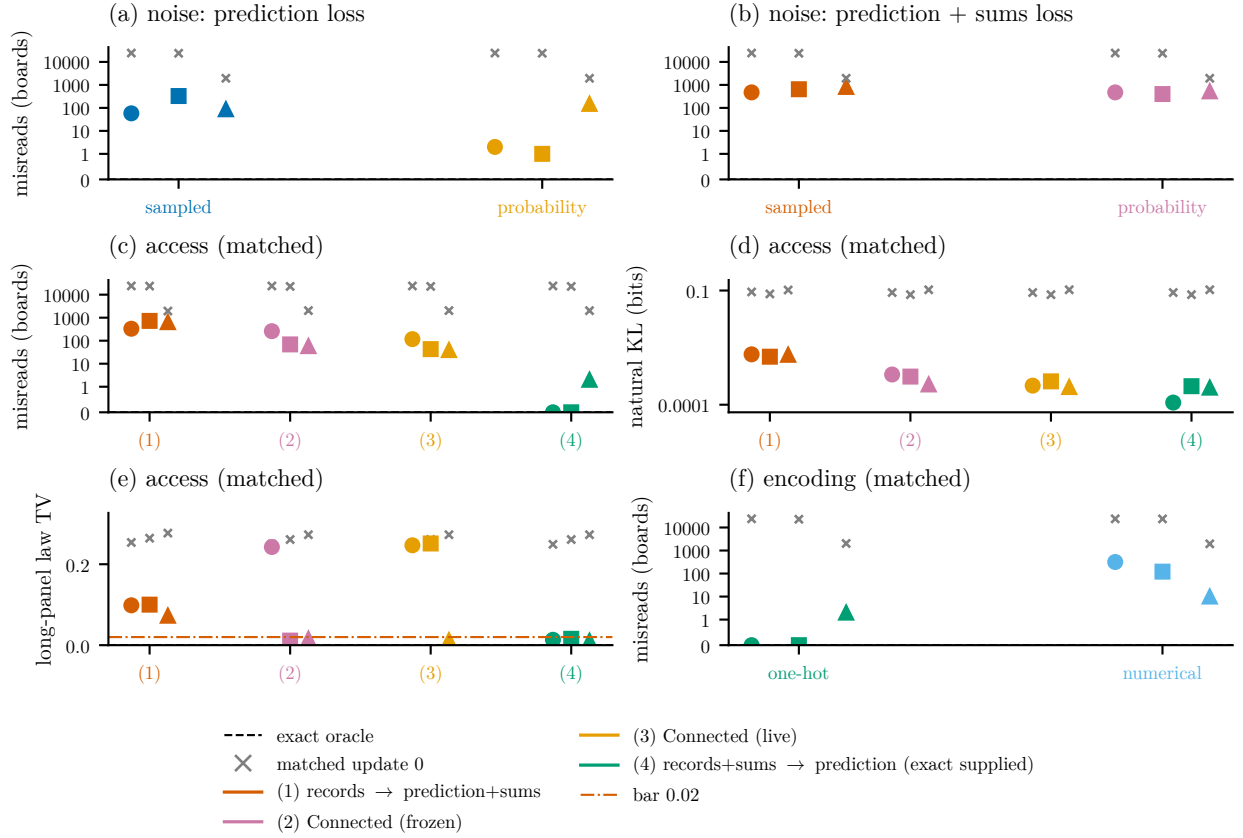


Figure 4: Target noise, access and encoding affect measured errors without guaranteeing prediction closure. Matched r13 comparisons: targets within versions (a,b), numbered access variants (c–e), encodings of exact sums (f). Shapes: seeds 0/1/2; grey crosses: matched update 0; dashed line: zero-error oracle except in (d), whose 0-bit oracle lies outside the logarithmic axis. Categorical misreads: 24,435 boards, rendering 0; natural KL: 60,000 positions; maximum long-panel TV: 384 N3–N8 endpoints, bar 0.02. No pooled means or bands. Controls are matched within panels.

(uniform sampling) locates the board failures: 57 of 58, 289 of 328 and 87 of 87 categorical misreads lie in the cutoff band; the rest lie at slot-1 sums 9 and 10, just below it (fig. 3f).

One might expect training on long histories to remove the history failures. r12 added history coverage, and its long-panel errors are smaller than r11’s (a historical comparison: curriculum and loss weighting also changed). Even so, none of the nine r12 records $\rightarrow$ prediction runs passes (three board-sampling arms $\times$ three seeds). Long-panel maxima reach 0.172 and short-panel maxima range 0.020–0.066. We diagnosed 162 selected histories from these runs. 120 stay within the diagnostic 0.02-TV bar. Of the remaining 42, 20 are board-associated (all coincident with a misread of the current board), 3 are recurrence-associated, and 19 are unresolved. Misreads therefore do not account for every failure. Enriching training with boards at the cutoff values (r11) did not repair the misreads consistently across seeds (section B).

5.3 Target noise contributes but is not sufficient

Claim: the noise of sampled targets is one contributing condition. In this matched r13 comparison, training on the exact law instead of one sampled next category removes target noise

and changes nothing else. For records $\rightarrow$ prediction, census misreads go from 58 / 328 / 87 with sampled targets to 2 / 1 / 149 with probability targets: two seeds fall to near zero and the third gets worse (fig. 4a). For records $\rightarrow$ prediction+sums they go from 473 / 648 / 815 to 473 / 400 / 523 (fig. 4b). No run passes under either target. Removing target noise neither repairs every seed nor produces a pass.

5.4 Computing or recovering the sums does not guarantee access

Claim: a network can produce the sums largely correctly, or hold them in recoverable form, and still not deliver them to the prediction; access is a separate achievement.

Supplied versus learned sums (r12, matched versions). In r12, the matched comparison of the declared versions differs in three ways at once: whether exact sums are supplied, how accurate the sums are, and whether the sums are supervised. The records+sums $\rightarrow$ prediction networks, which receive exact sums, pass 2 of 3 runs with 0–1 misreads. The records $\rightarrow$ prediction+sums networks, which are trained to output the sums but do not feed them to the prediction, pass 0 of 3 with 473–815 misreads. Among boards where their own sums output gets the slot-1 category right, their prediction still has a categorical misread on 326 / 617 / 770. Their sums output is correct on 19,269 / 20,516 / 19,544 of the 24,435 completed sum boards, so these networks compute the sums largely but not exactly. The comparison supports more reliable prediction with exact supplied sums under these conditions; it does not isolate access.

Connection (r13, matched). The direct test of access is r13’s connection experiment, which connects the learned sums output to the prediction and changes nothing else (fig. 4c–e). Relative to the disconnected networks (331 / 725 / 637 misreads), Connected (frozen) has 259 / 69 / 58 misreads, a 1.3–11-fold reduction, and Connected (live) has 117 / 43 / 39, a 2.8–17-fold reduction. Natural KL falls as well, from 0.0018–0.0021 bits to 0.0003–0.0006 bits. Access does not by itself produce reliable prediction: no connected run passes every criterion, and long-history failures can remain (Connected (live), seed 1: long-panel maximum 0.251). This supports access as a cause of the misread reduction, not a selective internal representation of the sums.

Swap test (r10). In r10, probes recover all five sums and the swap test passed calibration, but on the trained networks the tested alignment was not selective: 188–436 of 1,024 swaps changed another slot’s decoded sum. The failed selectivity means only that this test cannot support use under this alignment.

5.5 The form of access matters

Claim: the same exact information can be easy or hard to use, depending on how it is encoded. Given the same exact sums, the one-hot encoding gives 0 / 0 / 2 misreads and the numerical encoding ($s/36$, $1 - s/36$) gives 317 / 121 / 10 (fig. 4f). The two conditions differ only in the encoding. A candidate reason, which these runs do not test, is that a numerical input requires the prediction to place a threshold between adjacent values such as 12 and 13 twelfths, whereas a one-hot input gives each value its own input unit. A supplementary probe analysis of the cutoff values is in section E. We do not infer a precision ceiling from these results.

5.6 Prediction updates erode an exact sums computation

Claim: a computation that is exact can be lost when training continues on the prediction alone; preservation is a separate achievement. In r11, training continued from an exact sums network. With the prediction loss only, board-vector exactness on a fixed panel of 128 boards falls from 1.0 at update 0 to 13/128, 2/128 and 11/128 at update 1 and remains severely degraded at

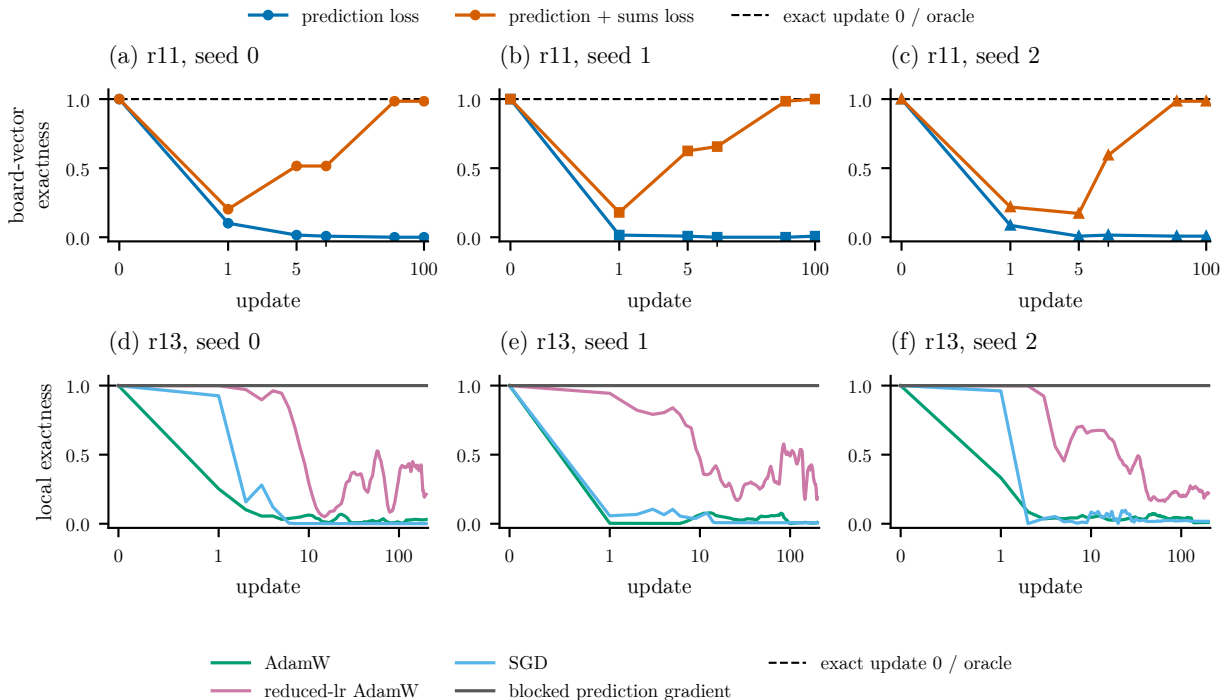


Figure 5: Prediction gradients erode exact sums at update 1, not uniquely under AdamW. Columns: seeds 0–2. (a–c) Compare r11 losses: exact starts on 24,435 boards; updates 1/5/10/50/100 use a fixed 128-board subset. Symlog separates 0/1. (d–f) r13, all 1,555 local histories: every tested active optimizer loses exactness at update 1. Update-1 correct counts, seeds 0/1/2, each out of 1,555: AdamW: 393 / 5 / 518; reduced-lr AdamW: 1,554 / 1,468 / 1,553; SGD: 1,440 / 89 / 1,495; blocked prediction gradient: 1,555 / 1,555 / 1,555. Blocked prediction gradient is a separate weight-decay control; dashed line: exact start/oracle. Movement-matched points: fig. D.2. r11/r13 is historical; no pooling or bands.

later sampled readouts (fig. 5a–c). Keeping the sums loss largely protects the computation, though not everywhere: at the endpoint, 24,434 / 24,435 / 24,419 of the 24,435 completed sum boards are correct.

r13 measured the first 200 updates exhaustively on the 1,555 local histories (fig. 5d–f). Exactness is lost at update 1 under every tested active optimizer, with very different severity. After one update, AdamW keeps 393 / 5 / 518 histories exact, AdamW with a tenfold smaller learning rate keeps 1,554 / 1,468 / 1,553, and SGD keeps 1,440 / 89 / 1,495. When the prediction gradient to the sums network is blocked and weight decay continues, all 1,555 histories stay exact through 200 updates in every seed. Within the range of cumulative movement that all three optimizers covered, exactness falls for each (fig. D.2). Erosion is therefore not specific to Adam. These runs do not separate the direction of the prediction gradient, the size of the displacement and the margins of the sums answers as causes.

5.7 Correct sums do not guarantee reliable prediction

Claim: even with exact sums supplied, reliable prediction is a further achievement that some seeds miss. With exact sums supplied as input, r8’s records+sums $\rightarrow$ prediction passes 3 of 3 runs, and records+sums $\rightarrow$ prediction (sums route only) passes 2 of 3. r12’s records+sums $\rightarrow$ prediction passes 2 of 3, and r13’s one-hot exact condition passes 1 of 3 (table B.2). In the

all-slots task (r10), with its own criteria, the version given exact sums passes 0 of 3. These are historical comparisons across studies with different training protocols. A pass is also not exhaustive correctness: the passing r13 seed still misreads 2 boards in the census. The failed criteria of the supplied-sums networks are specific: r12 seed 2 exceeds the bar on the reset swap (TV 0.0590), r13 seed 0 on the single L history (law TV 0.02275), and r13 seed 1 on the neutral swap (TV 0.02067). The sums are exact in all of them; candidates include the history update, the probability readout and the response to interventions. We have not identified their mechanisms.

6 Discussion

What we learned. Training a network on an upper theory gave it the lower theory only in a jagged form, and that jagged form is where its jagged competence came from. The network did learn something of A: the part of A that B depends on became readable, while the parts B ignores became less readable. But what it learned was uneven in the places that matter for B. Category misreads concentrated near the slot-1 cutoffs, where B’s categories change, even though a probe could read the slot-1 sum on 97–98% of boards; when A was trained as an explicit output it was largely but not exactly right; and part of the prediction’s failure sat where this partial A failed. Seen this way, jagged competence is not noise on top of an otherwise correct solution. Uneven acquisition, access and preservation of A explain part of B’s jagged competence; some failures remain unexplained. When B instead needed all five sums (the all-slots task), all five became readable, which suggests that the jagged form of A follows what B asks of it; that comparison is historical and crosses tasks.

Knowing an intermediate is not one thing. Our second lesson is that “the network knows A” hides five separate facts. A can be readable without established exact computation, computed without reaching the prediction, delivered in a form that supports less reliable prediction, or lost under further training, and an exact, delivered A can still leave failures in B. Matched comparisons isolate the effects of connection, encoding and prediction gradients: connecting a learned A to B cut misreads by up to 17-fold, the same exact A gave fewer misreads as one-hot inputs than as numerical inputs, and a single B update made an exact A inexact unless the B gradient was blocked. Having the right information is therefore not enough; how it reaches the decision, in what form, and whether it survives training all matter. Nor is more supervision of the intermediate automatically better: networks also trained to output the sums had more misreads (473–815) than networks trained on the prediction alone (58–328) in r13’s sampled-target condition (section 5.3).

Jaggedness has locatable sources. The known layers let us distinguish several sources of failure: missing access, missing preservation, a less usable encoding, noisy targets, and history coverage, assessed in a historical comparison. Some cases remain unexplained: 19 of the 42 diagnosed r12 histories that exceed the diagnostic bar, and the residual failures with exact sums. Average error and probe accuracy each looked healthy while these failures were present, so neither alone establishes complete, reliable realization of the task. “Jagged” names the phenomenon; this account, not the word, is the explanation.

Relation to other accounts. Coverage-based accounts explain jaggedness by gaps in what a system has seen [8]; history coverage is one such factor here, but the access, preservation and encoding effects appear in matched comparisons with identical training data, so coverage alone does not account for them. Other studies report gaps between what probes read and what outputs do [1, 3]; our connection experiment shows one way such a gap can cause errors and be reduced.

What we expect to carry over. We expect the two measurement cautions to hold wherever the same measurements are used: low average error does not establish reliability on constructed histories, and recoverability does not establish predictive use. Whether learning an upper theory yields a jagged lower theory in larger systems is a test to run, for example by checking which parts of a known intermediate become readable under final-output training, whether a learned intermediate output is used by the main prediction, and whether downstream training preserves it.

7 Limitations

1. *Scope.* One task family, small recurrent networks (about 40,000 parameters), three seeds per condition and fixed budgets: we report observed counts, not population frequencies, and cannot claim that the separations hold for other tasks, architectures or longer training.
2. *Constructed tests.* The law panels and the census locate failures but do not estimate how often they occur under the natural distribution.
3. *Confounded comparisons.* Comparisons across studies (r11 versus r12; anything involving r10) change more than one factor and are not attributed to a single cause; within r12, the version comparison changes supplied sums, sums accuracy and sums supervision together and does not isolate access.
4. *Internal structure.* Probe declines show reduced measured readability, not removal of information, for the tested probe families and support; probe boards are disjoint between fitting and evaluation but not necessarily unseen during network training, and rare r10 sums (28–36) lie outside the registered scope. The connection experiment changes the route, not the internal state, and the failed swap selectivity excludes no other representation. None of these establishes how the free networks represent or use the sums internally.
5. *Encoding.* The encoding experiment compares two encodings and does not establish a precision ceiling; the supplementary cutoff discrimination is established on that support, not across all cutoff boards, and does not diagnose the numerical-encoding failures.
6. *Mechanisms.* Erosion depends on the prediction gradient, but the runs do not separate gradient direction, displacement and answer margins, so interference is not established; the residual failures of networks given exact sums are located but not explained, and a pass is not exhaustive correctness.

8 Conclusion

We asked where jagged competence comes from. In a task built from two known layers, the answer was that a network trained on the upper layer learns the lower one only in a jagged form, readable where the upper layer needs it, with B’s category responses failing near the cutoffs, and not reliably delivered or kept; these separations explain part of its failures, and some remain unexplained. Knowing an intermediate quantity turned out to be several separate achievements, each of which can fail on its own, and average error and probes each hid the failures.

The practical lesson is that a network’s competence on a task built in layers should not be judged by its average error or by what a probe can read, but by measuring, layer by layer, what it has acquired, delivered and kept. Whether learning an upper theory yields a jagged lower theory in larger systems is a test to run.

A note on the Six Birds programme

Six Birds’ organizing picture motivates constructing upper computations over lower-level objects [26, §1.1]. Whether final-output training discovers such a construction is this programme’s empirical hypothesis, not a theorem about trained networks. Foundations I separates stability, novelty and directionality certificates and keeps them independent. Our five achievements are different empirical measurements. Read in the theory’s terms, our central finding is that learning a strict extension induced the lower theory only partially, and these separations help explain part of the measured jagged competence; this is an empirical observation in one task, not a test of the theory’s formal certificate claims, and it does not establish that the networks implement those layers internally.

Competing interests

The author also developed Six Birds Theory, which motivates the task’s layered design and learning hypothesis; the empirical results stand independently of the theory and bear on that application without validating its formal claims. The author declares no other competing interests.

AI-assistance disclosure

The author used AI tools in preparing this work and takes full responsibility for its content.

Availability of materials

The source package contains the manuscript, the claims ledger and evidence manifest, the scripts that rebuild every number, figure and table, and the released data files with their schema (section F). The study code is available at <https://github.com/ioannist/six-birds-ml>.

References

- [1] Sanjay Basu, Sadiq Y. Patel, Parth Sheth, Bhairavi Muralidharan, Namrata Elamaram, Aakriti Kinra, John Morgan, and Rajaie Batniji. Interpretability without Actionability: Mechanistic Methods Cannot Correct Language Model Errors Despite Near-Perfect Internal Representations. *arXiv*, 2026. URL <https://arxiv.org/abs/2603.18353>.
- [2] Yonatan Belinkov. Probing Classifiers: Promises, Shortcomings, and Advances. *Computational Linguistics*, 2022. URL <https://arxiv.org/abs/2102.12452>.
- [3] Ishita Darade and Sushrut Thorat. Represented Is Not Computed: A Causal Test of Candidate Algorithmic Intermediates in a Transformer. *arXiv*, 2026. URL <https://arxiv.org/abs/2605.22488>.
- [4] Grégoire Delétang, Anian Ruoss, Jordi Grau-Moya, Tim Genewein, Li Kevin Wenliang, Elliot Catt, Chris Cundy, Marcus Hutter, Shane Legg, Joel Veness, and Pedro A. Ortega. Neural Networks and the Chomsky Hierarchy. *International Conference on Learning Representations*, 2023. URL <https://arxiv.org/abs/2207.02098>.
- [5] Fabrizio Dell’Acqua, Edward McFowland III, Ethan Mollick, Hila Lifshitz, Katherine C. Kellogg, Saran Rajendran, Lisa Kraye, François Cabelon, and Karim R. Lakhani. Navigating the Jagged

- Technological Frontier: Field Experimental Evidence of the Effects of Artificial Intelligence on Knowledge Worker Productivity and Quality. *Organization Science*, 37(2):403–423, 2026. doi: 10.1287/orsc.2025.21838.
- [6] Yanai Elazar, Shauli Ravfogel, Alon Jacovi, and Yoav Goldberg. Amnesic Probing: Behavioral Explanation with Amnesic Counterfactuals. *Transactions of the Association for Computational Linguistics*, 2021. URL <https://arxiv.org/abs/2006.00995>.
 - [7] Nelson Elhage, Tristan Hume, Catherine Olsson, Nicholas Schiefer, Tom Henighan, Shauna Kravec, Zac Hatfield-Dodds, Robert Lasenby, Dawn Drain, Carol Chen, Roger Grosse, Sam McCandlish, Jared Kaplan, Dario Amodei, Martin Wattenberg, and Christopher Olah. Toy Models of Superposition. *arXiv*, 2022. URL <https://arxiv.org/abs/2209.10652>.
 - [8] Joshua Gans. A Model of Artificial Jagged Intelligence. *arXiv*, 2026. URL <https://arxiv.org/abs/2601.07573>.
 - [9] Atticus Geiger, Hanson Lu, Thomas Icard, and Christopher Potts. Causal Abstractions of Neural Networks. *Advances in Neural Information Processing Systems*, 2021. URL <https://arxiv.org/abs/2106.02997>.
 - [10] Atticus Geiger, Zhengxuan Wu, Christopher Potts, Thomas Icard, and Noah D. Goodman. Finding Alignments Between Interpretable Causal Variables and Distributed Neural Representations. *Conference on Causal Learning and Reasoning*, 2024. URL <https://arxiv.org/abs/2303.02536>.
 - [11] Robert Geirhos, Jörn-Henrik Jacobsen, Claudio Michaelis, Richard Zemel, Wieland Brendel, Matthias Bethge, and Felix A. Wichmann. Shortcut Learning in Deep Neural Networks. *Nature Machine Intelligence*, 2020. URL <https://arxiv.org/abs/2004.07780>.
 - [12] Katherine L. Hermann and Andrew K. Lampinen. What shapes feature representations? Exploring datasets, architectures, and training. *Advances in Neural Information Processing Systems*, 2020. URL <https://arxiv.org/abs/2006.12433>.
 - [13] John Hewitt and Percy Liang. Designing and Interpreting Probes with Control Tasks. *EMNLP-IJCNLP*, 2019. URL <https://arxiv.org/abs/1909.03368>.
 - [14] Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. Distilling the Knowledge in a Neural Network. *arXiv*, 2015. URL <https://arxiv.org/abs/1503.02531>.
 - [15] Adam Tauman Kalai and Santosh S. Vempala. Calibrated Language Models Must Hallucinate. *arXiv*, 2024. URL <https://arxiv.org/abs/2311.14648>.
 - [16] James Kirkpatrick, Razvan Pascanu, Neil Rabinowitz, Joel Veness, Guillaume Desjardins, Andrei A. Rusu, Kieran Milan, John Quan, Tiago Ramalho, Agnieszka Grabska-Barwinska, Demis Hassabis, Claudia Clopath, Dharshan Kumaran, and Raia Hadsell. Overcoming catastrophic forgetting in neural networks. *Proceedings of the National Academy of Sciences*, 2017. URL <https://arxiv.org/abs/1612.00796>.
 - [17] Kenneth Li, Aspen K. Hopkins, David Bau, Fernanda Viégas, Hanspeter Pfister, and Martin Wattenberg. Emergent World Representations: Exploring a Sequence Model Trained on a Synthetic Task. *International Conference on Learning Representations*, 2023. URL <https://arxiv.org/abs/2210.13382>.

- [18] Michael McCloskey and Neal J. Cohen. Catastrophic Interference in Connectionist Networks: The Sequential Learning Problem. In *Psychology of Learning and Motivation*, volume 24, pages 109–165. 1989. doi: 10.1016/S0079-7421(08)60536-8. No arXiv identifier recorded.
- [19] Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and Editing Factual Associations in GPT. *Advances in Neural Information Processing Systems*, 2022. URL <https://arxiv.org/abs/2202.05262>.
- [20] Neel Nanda, Lawrence Chan, Tom Lieberum, Jess Smith, and Jacob Steinhardt. Progress measures for grokking via mechanistic interpretability. *International Conference on Learning Representations*, 2023. URL <https://arxiv.org/abs/2301.05217>.
- [21] Mohammad Pezeshki, Sékou-Oumar Kaba, Yoshua Bengio, Aaron Courville, Doina Precup, and Guillaume Lajoie. Gradient Starvation: A Learning Proclivity in Neural Networks. *Advances in Neural Information Processing Systems*, 2021. URL <https://arxiv.org/abs/2011.09468>.
- [22] Abhilasha Ravichander, Yonatan Belinkov, and Eduard Hovy. Probing the Probing Paradigm: Does Probing Accuracy Entail Task Relevance? *EACL*, 2021. URL <https://arxiv.org/abs/2005.00719>.
- [23] Harshay Shah, Kaustav Tamuly, Aditi Raghunathan, Prateek Jain, and Praneeth Netrapalli. The Pitfalls of Simplicity Bias in Neural Networks. *Advances in Neural Information Processing Systems*, 2020. URL <https://arxiv.org/abs/2006.07710>.
- [24] Seine A. Shintani. Arithmetic OOD Failure Unfolds in Stages in Minimal GPTs. *arXiv*, 2026. URL <https://arxiv.org/abs/2603.26828>.
- [25] Wesley Shu and Peng Wei. Artificial Jagged Intelligence as Uneven Optimization Energy Allocation: Capability Concentration, Redistribution, and Optimization Governance. *arXiv*, 2026. URL <https://arxiv.org/abs/2605.01420>.
- [26] Ioannis Tsiokos. Six Birds: Foundations of Emergence Calculus. *arXiv*, 2026. doi: 10.48550/arXiv.2602.00134. URL <https://arxiv.org/abs/2602.00134>.
- [27] Ioannis Tsiokos. Six Birds Foundations III: A Finite Audited Interaction Calculus for SBT, 2026. URL <https://doi.org/10.5281/zenodo.23096391>.
- [28] Ioannis Tsiokos. Strict Theory Extension on a Lawful Continuous Cantor Shell, 2026. URL <https://doi.org/10.5281/zenodo.19189353>. Section 3.5, Definability as factorization; Section 6, Theorem 3. The present study uses non-factorization only for the task observable, not as evidence of an internal network layer.
- [29] Keyon Vafa, Justin Y. Chen, Ashesh Rambachan, Jon Kleinberg, and Sendhil Mullainathan. Evaluating the World Model Implicit in a Generative Model. *Advances in Neural Information Processing Systems*, 2024. URL <https://arxiv.org/abs/2406.03689>.
- [30] Tian Ye, Zicheng Xu, Yuanzhi Li, and Zeyuan Allen-Zhu. Physics of Language Models: Part 2.1, Grade-School Math and the Hidden Reasoning Process. *International Conference on Learning Representations*, 2025. URL <https://arxiv.org/abs/2407.20311>.
- [31] Tianhe Yu, Saurabh Kumar, Abhishek Gupta, Sergey Levine, Karol Hausman, and Chelsea Finn. Gradient Surgery for Multi-Task Learning. *Advances in Neural Information Processing Systems*, 2020. URL <https://arxiv.org/abs/2001.06782>.

A Study history

Table A.1: Analytical appendix A. Exact recorded counts or decisions; no pooled reader accuracies. Complete support, settings, controls and per-seed records are in the numbered supplementary tables and released data.

study	recorded compute	version/review records
r8	CPU-era; processor model not recorded in saved per-run summaries	11
r9	CPU-era; processor model not recorded in saved per-run summaries	1
r10	GPU 3 / CUDA fp32 execution recorded in summaries and host verification; GPU model and driver not recorded in the public study artifacts	20
r11	GPU 3 / CUDA fp32 execution recorded in summaries and host verification; GPU model and driver not recorded in the public study artifacts	13
r12	GPU 3 / CUDA fp32 execution recorded in summaries and host verification; GPU model and driver not recorded in the public study artifacts	14
r13	GPU 3 / CUDA fp32 execution recorded in summaries and host verification; GPU model and driver not recorded in the public study artifacts	44

The evidence history is retained in Supplement Table S3. Sampling, training and criteria for r8–r13 are in Supplement Tables S4, S6, S8, S10, S12 and S14; their complete endpoint matrices are S5, S7, S9, S11, S13 and S15. A corrected audit does not change the registered training endpoint. Follow-ups and supplementary panels have their own scope; comparisons between studies are historical.

B Analytical study comparison

Table B.1: Analytical appendix B. Exact recorded counts or decisions; no pooled reader accuracies. Complete support, settings, controls and per-seed records are in the numbered supplementary tables and released data.

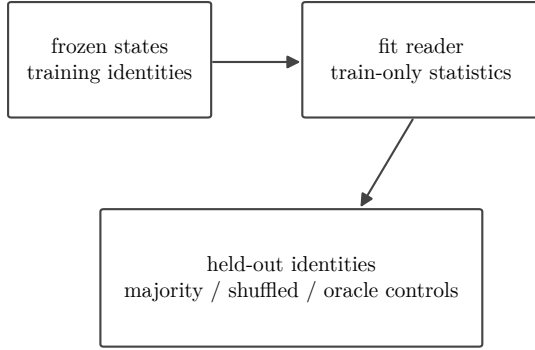
study	registered passes	original-law runs	law TV bar / support
r8	5	9	0.02 (576 history endpoints)
r10	0	9	0.05 (180 law rows)
r11	1	24	0.02 (576 history endpoints)
r12	2	15	0.02 (576 history endpoints)
r13	1	27	0.02 (576 history endpoints)

Table B.2: Access does not guarantee registered reliability. Each vertically ordered triple is seeds 0/1/2, never a pooled mean. Compare within study; cross-study comparisons are historical. r8 full history maximum, r12/r13 long N3–N8 maximum: bar 0.02. r10 is a different task: 180 law rows, original bar 0.05 (equal bar 0.02). Natural panels: 5,000 episodes x 12 rounds. Complete per-seed failed bars, census support and labels: Supplement Table S2.

condition	natural KL (bits)	law TV (maximum)	B passes
r8: records → prediction	0.00029829	0.25	0/3
	7.047e-05	0.2548	
	0.00021024	0.25649	
r8: records+sums → prediction (sums route only)	0.00033865	0.017721	2/3
	6.7874e-05	0.011403	
	6.5306e-05	0.020775	
r8: records+sums → prediction	0.00025483	0.018665	3/3
	6.8052e-05	0.0051108	
	8.0944e-05	0.0059933	
r10: records → prediction	0.016034	0.3219	0/3
	0.019138	0.28963	
	0.016491	0.30706	
r10: records → prediction+sums	0.032325	0.27737	0/3
	0.033959	0.25519	
	0.035928	0.23295	
r10: records+sums → prediction (sums route only)	0.0070603	0.12242	0/3
	0.0059933	0.079414	
	0.0057117	0.10517	
r12: records → prediction+sums	0.0015898	0.064965	0/3
	0.0020044	0.087776	
	0.0022981	0.044799	
r12: records+sums → prediction	0.00011363	0.014557	2/3
	0.00029505	0.010587	
	0.00025691	0.012218	
r13: records → prediction+sums	0.0021068	0.0985	0/3
	0.0018252	0.10016	
	0.0020459	0.072752	
r13: Connected (frozen)	0.00062178	0.24257	0/3
	0.0005485	0.011304	
	0.00034049	0.015699	
r13: Connected (live)	0.00031731	0.24683	0/3
	0.00041094	0.25128	
	0.00029062	0.011997	
r13: records+sums → prediction (exact supplied)	0.00011384	0.013305	1/3
	0.00030667	0.01569	
	0.00028061	0.010661	
r13: records+sums → prediction (Numerical encoding)	0.00083668	0.24067	0/3
	0.00079477	0.069823	
	0.00054318	0.18902	
r8 board-only calibration	0.013514	0.1875	0/1
exact task oracle	0	0	ceiling

Supplement Tables S16–S19 retain all census counts, sampled support, law cases and r10 exposure. Categorical misreads and continuous TV errors remain separate. The r10 all-slots task uses 180 law rows with its own original-law 0.05 bar; the original task uses 576 constructed history endpoints and a 0.02 bar. Equal-law runs are controls, not evidence for the original-law pathway claim.

(a) Frozen-state probe



(b) Selective swap test

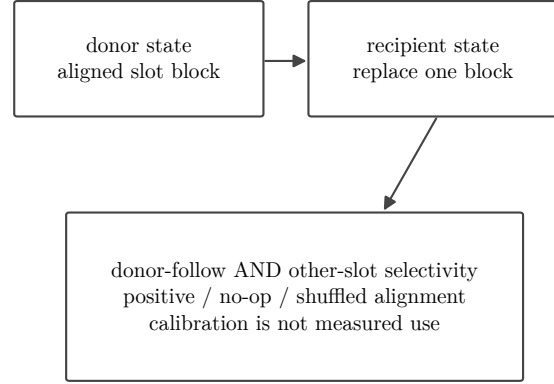


Figure C.1: Separate probe (a) and swap-test (b) procedures. Readers and alignment fit on training identities only; scoring identities are held out. Positive, no-op and shuffled controls calibrate the procedure, while donor-follow and selectivity concern the evaluated carrier. This schematic has no empirical seed average or accuracy; full controls and support are in Appendix C.

C Calibration and swap procedures

Table C.1: Analytical appendix C. Exact recorded counts or decisions; no pooled reader accuracies. Complete support, settings, controls and per-seed records are in the numbered supplementary tables and released data.

study	law	positive / null	deciding quantity	pass	support	measured value
r8	original	current_board_	B deciding criteria	no	576	0.1875
r8	original	null forced_A	B deciding criteria	yes	576	0
r8	original	forced_raw	B deciding criteria	yes	576	0
r10	original	exact_oracle	B deciding criteria	yes	180 law rows	4.4703e-08
r10	original	mode_blind	B deciding criteria	no	180 law rows	0.48635
r10	original	untrained	B deciding criteria	no	180 law rows	0.42896
r10	original	slot 1: hard / rotated / no-op / shuffled	donor-follow and selectivity	yes	1,024	1/1/0/0.1123
r10	original	slot 2: hard / rotated / no-op / shuffled	donor-follow and selectivity	yes	1,024	1/1/0/0.053711
r10	original	slot 3: hard / rotated / no-op / shuffled	donor-follow and selectivity	yes	1,024	1/1/0/0
r10	original	slot 4: hard / rotated / no-op / shuffled	donor-follow and selectivity	yes	1,024	1/1/0/0.18164
r10	original	slot 5: hard / rotated / no-op / shuffled	donor-follow and selectivity	yes	1,024	1/1/0/0.18164

Continued on next page

study	law	positive / null	deciding quantity	pass	support	measured value
r10	original	planted contamination	other-slot changes detected	yes	1,024	949/950/926/968/959
r11	original	current_board_null	B deciding criteria	no	576	0.1875
r11	original	oracle_A	B deciding criteria	yes	576	0
r11	original	oracle_raw	B deciding criteria	yes	576	0
r12	original	current_board_null	B deciding criteria	no	576	0.1875
r12	original	forced_A_oracle	B deciding criteria	yes	576	0
r12	original	oracle	B deciding criteria	yes	576	0
r13	original	board_only	B deciding criteria	no	576 history endpoints	0.1875
r13	original	oracle	B deciding criteria	yes	576 history endpoints	0

Deciding-function positive/null comparisons are Supplement Table S20. Table S21 summarizes fixed-pair support, positive validity and other-slot changes; its named detailed file retains per-slot negative calibration, selectivity and intervals. Success of an oracle calibration is not evidence of selective use by a trained carrier.

C.1 Measurement details

Prediction criteria. Unseen KL is computed on test positions whose preceding pair of boards never occurred at a training position; its support is recorded per run (about 20,000–26,000 positions). Witness recovery is $1 - \bar{K}/J$, where $\bar{K}$ is the mean KL in bits on the 64 histories made of an anchor and one N board, and $J = 0.061278 \dots$ bits is the Jensen–Shannon divergence between the two laws, the mean KL of a predictor that ignores the anchor. The state-intervention panels reset and set the mode, hold it through neutral boards, substitute a board of the same category and exchange upper states. Under the equal law, witness recovery is undefined and is replaced by a check that the two witness predictions agree within 0.02 TV. r10 uses its own conjunction, including bars derived from its calibrated history-blind predictor and law TV at most 0.05 under its original law; the complete study-specific criteria for every study, including r10’s covered-value KL, equal-law TV and witness-separation rules, are listed in Supplement Tables S4, S8, S10, S12 and S14 (r8, r10, r11, r12, r13; r9 has no pass criteria).

Census. The two contexts are produced by executing an L and an H anchor through the network, so a misread can also reflect a failure to establish the mode. The signature distance is the summed absolute difference over the two contexts. From r11 on, the census covers all 24,435 boards in each of four fixed renderings and counts are reported per rendering; r9 used one rendering per board.

Probes. In the original task the upper state is probed after one board from a zero state; in r10 it is probed after recurrent histories. The 2,048 probe-training boards and the 512 evaluation boards per slot are disjoint from each other but not necessarily unseen during network training. A positive gain is supported when the lower bound of its paired 95% interval (1,000 resamples of the same 512 boards) is above zero and, for linear probes, both fits converged; an interval wholly below zero

supports a decline. Linear probes use balanced class weights, which can push ordinary accuracy below the majority floor.

Cutoff discrimination (supplementary). A separate analysis of the probability-target records $\rightarrow$ prediction networks shows that probes separate the sums 12 from 13 (150 evaluation pairs) and 23 from 24 (13 pairs) in every seed, while those networks still fail registered prediction tests. This establishes discrimination on that support, not across all cutoff boards, and does not diagnose the numerical-encoding failures.

Swap test. For the networks that learn from records, the alignment is a mean-centred orthogonal map, fitted on training data only, from the 16-dimensional raw carrier onto 15 coordinates that track the exact sums (three per slot) plus one residual coordinate; the patched state is mapped back through its inverse. It is a fixed, fitted alignment, not a searched subspace. The trained positive (records+sums $\rightarrow$ prediction) is instead patched by replacing the queried slot’s hard sum answer directly. The r10 swap test uses a fixed panel of 1,024 board pairs per slot and requires at least 20 pairs per slot on which the network is accurate for both boards. Donor-follow is the fraction of those accurate pairs for which the patched prediction is closer in TV to the donor’s law than to the base board’s law. Selectivity requires that an independent decoder finds no change in the other four slots’ sums. Four checks are reported separately: whether the trained positive (at the 20,000-update endpoint) follows the donor, whether the no-op and shuffled-alignment negatives do not, whether the swap is selective, and whether the evaluated networks follow the donor.

D Learning curves and first-update erosion

Table D.1: Analytical appendix D. Exact recorded counts or decisions; no pooled reader accuracies. Complete support, settings, controls and per-seed records are in the numbered supplementary tables and released data.

update-1 condition	correct counts (seeds 0/1/2)	histories per seed
AdamW	393 / 5 / 518	1,555
reduced-lr AdamW	1554 / 1468 / 1553	1,555
SGD	1440 / 89 / 1495	1,555
blocked prediction gradient	1555 / 1555 / 1555	1,555

All per-update local counts and movements are in Supplement Table S27. The supplement also contains every study’s learning curves, without checkpoint selection. The figures below show why low natural KL is not a sufficient reliability measurement, and the erosion of the sums computation against cumulative movement.

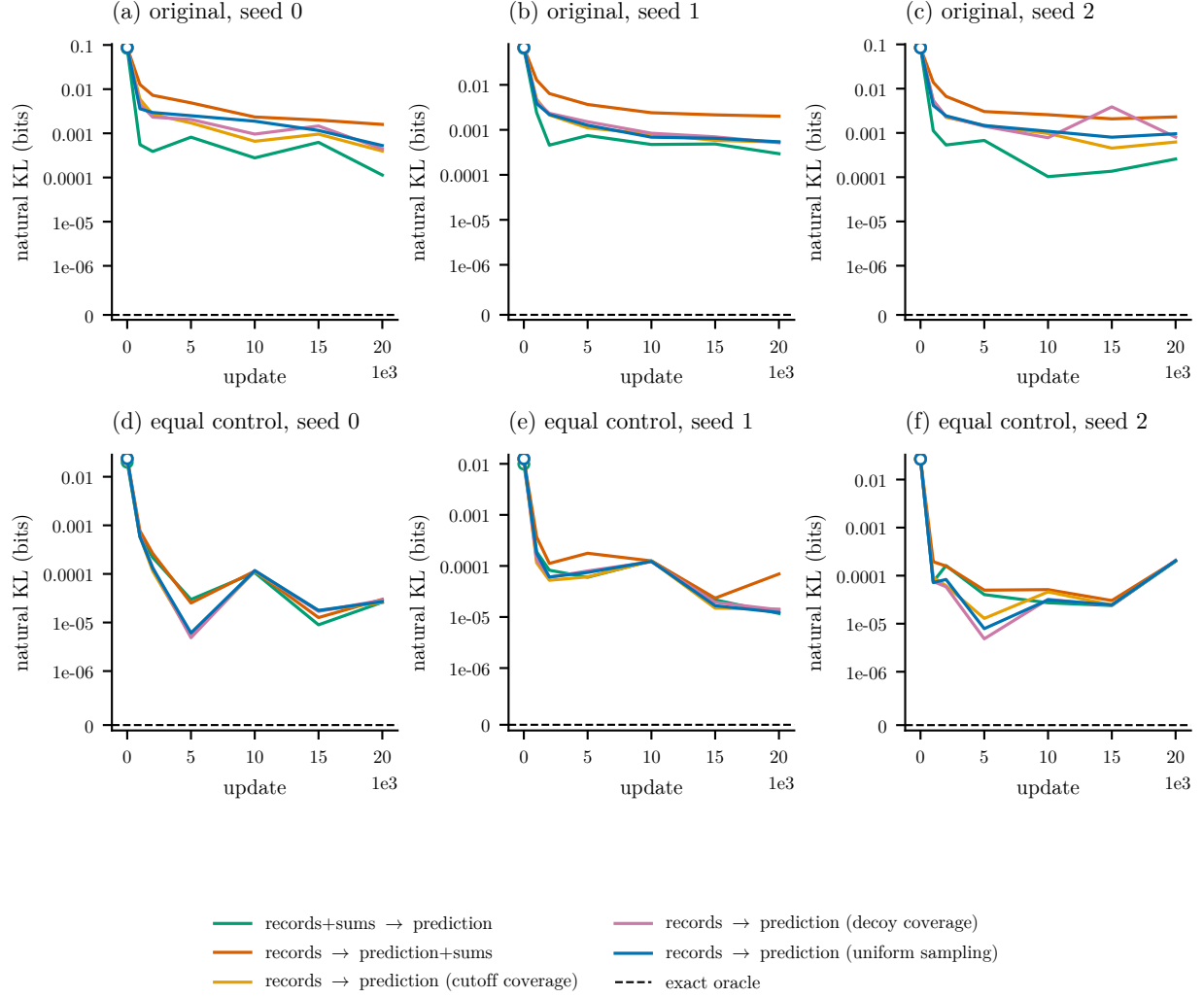


Figure D.1: Recorded r12 learning curves show changes in average error, not sufficient evidence of reliable competence. Every seed and condition is retained; (a–c) original and (d–f) equal laws are separate rows, in seed order 0–2. Open points mark each condition’s own update-0 control; dashed zero is the exact oracle. The quantity is natural KL in bits on 5,000 natural episodes of 12 rounds (60,000 scored positions), not a loss or a census frequency. No smoothing, selected checkpoint or across-seed band is used. Compare conditions within the same seed and law only; cross-study comparisons are historical. Natural KL alone does not establish reliability.

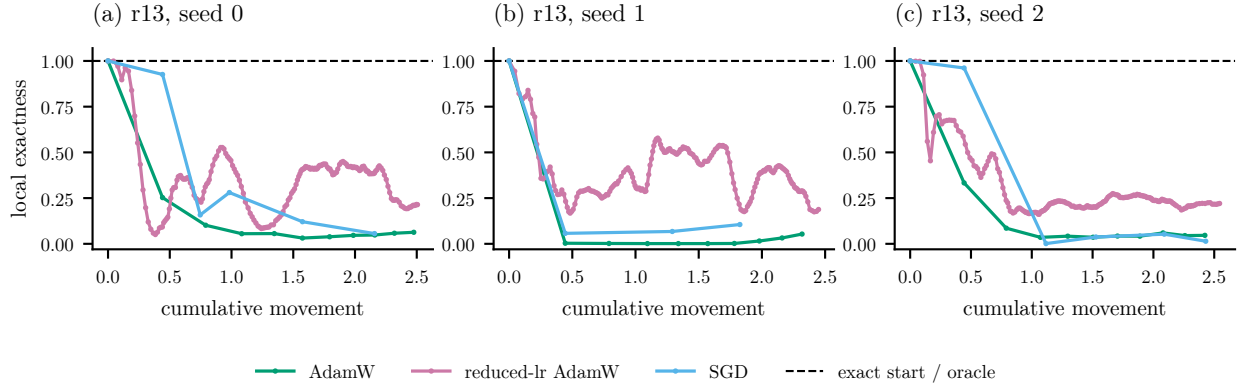


Figure D.2: Exact sums erode across active optimizers at common recorded movement. (a–c) Seeds 0–2, all 1,555 local histories. Compare AdamW, reduced-rate AdamW and SGD within each seed’s three-active-arm common range; points are observed, not interpolated. Blocked prediction gradient is excluded from that range and remains a separate control in fig. 5. Dashed line: exact start/oracle. No pooled means or confidence bands.

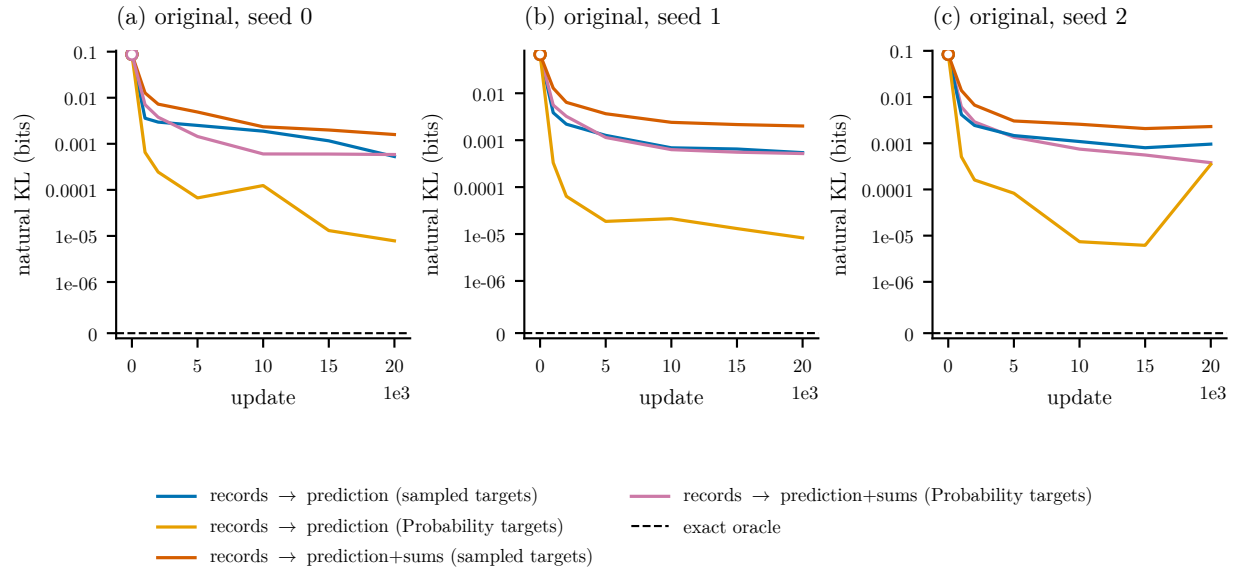


Figure D.3: Recorded r13 noise learning curves show changes in average error, not sufficient evidence of reliable competence. Every seed and condition is retained; (a–c) original; no equal-law arms were run, in seed order 0–2. Open points mark each condition’s own update-0 control; dashed zero is the exact oracle. The quantity is natural KL in bits on 5,000 natural episodes of 12 rounds (60,000 scored positions), not a loss or a census frequency. No smoothing, selected checkpoint or across-seed band is used. Compare conditions within the same seed and law only; cross-study comparisons are historical. Natural KL alone does not establish reliability.

E Probes and geometry

Table E.1: Analytical appendix E. Exact recorded counts or decisions; no pooled reader accuracies. Complete support, settings, controls and per-seed records are in the numbered supplementary tables and released data.

raw linear slot-1 sums reader	seed	correct	boards	gain CI95	converged
r11 records → prediction+sums (sums dose 100%)	0	484	512	0.5332/0.62695	yes
r11 records → prediction+sums (sums dose 100%)	1	491	512	0.60938/0.69336	yes
r11 records → prediction+sums (sums dose 100%)	2	500	512	0.68555/0.76172	yes
r12 records → prediction+sums (sums dose 100%)	0	483	512	0.53315/0.62305	yes
r12 records → prediction+sums (sums dose 100%)	1	509	512	0.64648/0.72852	yes
r12 records → prediction+sums (sums dose 100%)	2	505	512	0.69722/0.77148	yes

Complete reader tables, floors, paired intervals and convergence are in Supplement Tables S22–S24. Original and supplementary geometry support are in S25–S26. The supplementary geometry figure compares the two recorded conditions of records → prediction, sampled and probability targets; its scope is stated in section C.1.

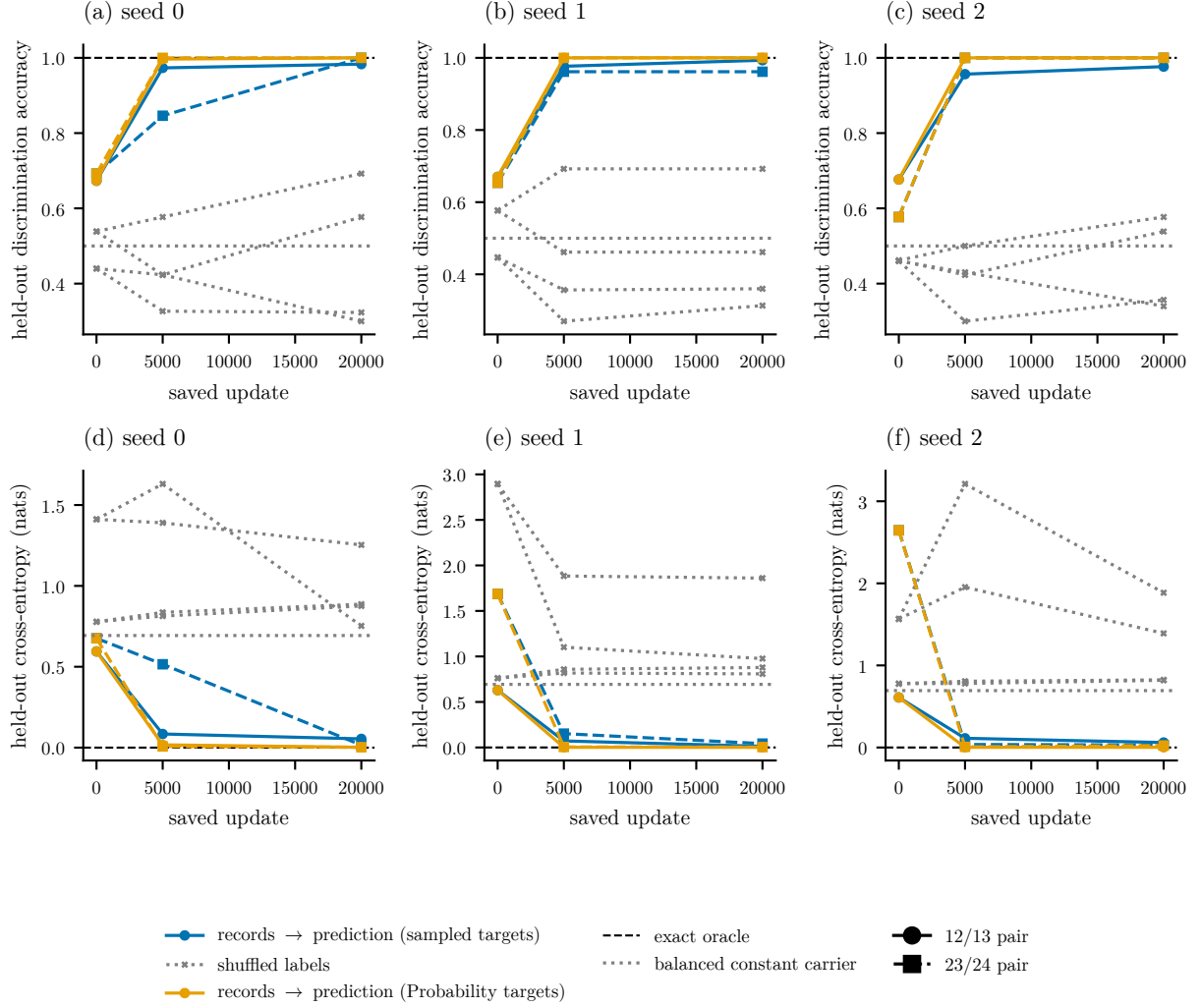


Figure E.1: Cutoff discrimination can succeed while prediction still fails. The supplementary panel was bound before states were read, with disjoint fit/evaluation identities and fixed renderings. Three seed columns show saved 0/5k/20k sampled- and probability-target prediction-loss checkpoints only, with no network updates: (a–c) accuracy and (d–f) cross-entropy, in seed order 0–2. Compare sampled- and probability-target readers for the same seed and pair; circles/solid lines are 12/13 (135 fitting pairs, 150 evaluation pairs; 300 endpoint labels), squares/dashed lines are 23/24 (12/13 pairs; 26 labels). Grey crosses are the saved shuffled-label floors; balanced constant floors are 0.5 accuracy and $\ln 2$ loss, and the exact ceiling is 1 accuracy / 0 loss. No confidence bands are implied. Distances and rendering variation remain descriptive in Appendix E’s support tables; successful discrimination is not proof of predictive use or of a precision ceiling.

F Compute and reproduction

Table F.1: Analytical appendix F. Exact recorded counts or decisions; no pooled reader accuracies. Complete support, settings, controls and per-seed records are in the numbered supplementary tables and released data.

study	condition (all seeds)	seconds (0/1/2)	execution scope	aggregation
r13	AdamW	21.962/19.645/20.531	200-update erosion; recorded execution	examples, not totals
r13	SGD	22.219/18.44/18.835	200-update erosion; recorded execution	examples, not totals
r13	blocked prediction gradient	19.329/17.844/17.231	200-update erosion; recorded execution	examples, not totals
r13	reduced-lr AdamW	19.241/19.387/20.737	200-update erosion; recorded execution	examples, not totals
r8	records $\rightarrow$ prediction	1855.1/1783.3/1895.1	CPU endpoint	examples, not totals
r8	records+sums $\rightarrow$ prediction	1765.2/1810.8/1837.4	CPU endpoint	examples, not totals
r8	records+sums $\rightarrow$ prediction (sums route only)	1670.6/1688.7/1682.7	CPU endpoint	examples, not totals

Supplement Table S28 summarizes recorded compute quantities; its named detailed file retains every recorded compute and allocation entry. Unknown historical execution metadata remain unknown. The public numerical package contains the manifest, ledger, hash-bound inputs and both TeX entry points. Rebuild both PDFs from the paper directory with `latexmk -pdf main.tex` and, separately,

`latexmk -pdf -jobname=supplement`

`supplement/supplement.tex`. The package check lists transitive inputs separately for each PDF and refuses overfull boxes exceeding 10 pt. Byte-identical asset rebuilds do not imply byte-identical TeX-engine PDFs, which contain build metadata.

F.1 Training details

Sums interface. The supplied sums enter the upper state as five 36-way one-hot vectors, projected to 32 dimensions by a bias-free linear map. The forward pass uses the hard one-hot answer; training gradients pass through the softmax behind it (a straight-through estimator). The exact sums network is a learned per-slot update (width 64) that is exact on all 1,555 local histories and is kept frozen except in the erosion continuation.

Episodes. In r8 and r11, training episodes have up to eight rounds and omit the third consecutive N board from both input and loss. From r12 on, each batch holds 128 untruncated nine-round natural episodes and 128 coverage prefixes (an L or H anchor followed by k N boards, $k = 1, \dots, 8$, eight prefixes per anchor and length). Coverage prefixes are scored only at their final position, and the prediction loss is split equally between natural positions and coverage endpoints.

Rarity and dose. The rarity conditions of r11 and r12 change only which board is drawn within a category (25% enrichment at the cutoff sums or at matched decoy sums), keeping the category draws and targets paired. The dose conditions of r11 apply the sums loss on nested 1%, 10% or 100% of rounds and divide by all active rounds times five slots.

Erosion replays. The r13 erosion runs last 200 updates and use r11’s truncated episodes, not history coverage. They differ only in how the sums network is updated: AdamW (learning rate 0.003); AdamW with learning rate 0.0003; SGD without momentum, whose fixed step size is chosen so that the norm of its gradient-only displacement on the first shared batch matches that of AdamW, with weight decay applied separately (multiplier 0.99997 per update); and a blocked condition in which the prediction gradient to the sums network is zero while AdamW weight decay continues. All other parameters use AdamW at 0.003 throughout.